Vivekanand Education Society's Institute of Technology

Department of Computer Engineering

Group No.: 52

Date :- 11/08/2025

Project Synopsis Template (2025-26) - Sem VII

SafeHer : ML-Powered Women Safety & Emergency System.

Name of The Mentor:-Mrs. Prerna Solanke
Assistant Professor,CMPN.

Riya Lasi	Mahek Kataria	Rahul Kithani	Manish Motwani
V.E.S.I.T	V.E.S.I.T	V.E.S.I.T	V.E.S.I.T
2022.riya.lasi@ves.ac.in	2022.mahek.kataria@ves.ac.in	2022.rahul.kithani@ves.ac.in	2022.manish.motwani@ves.ac.in

Abstract:-

Women's safety has become a pressing concern globally, with rising cases of harassment, assault, and other crimes against women. Despite technological advancements, there is still a lack of real-time, proactive safety measures that can predict, detect, and respond to potential threats before they escalate. This project, Ml-Powered Women Safety & Emergency System, is designed to address this gap by integrating predictive analytics, real-time GPS tracking, intelligent voice-based threat detection, and crime data analysis into one comprehensive solution.

The system leverages historical crime datasets, geospatial analysis, and Ml-based models to assess the risk level of any given location instantly. It also provides crime pattern analysis by age group, enabling more targeted safety recommendations for young women, minors, and elderly citizens. Using real-time GPS tracking, it can suggest alternate safe routes to avoid high-risk areas, functioning as a safety-oriented navigation assistant.

One of the most innovative aspects is voice-based threat detection. The system passively processes ambient audio to detect distress through emotion recognition, gender identification, speech-to-text conversion, and keyword monitoring, all without requiring the victim to manually operate the device. When combined with location-based crime risk maps, this feature allows the system to send instant alerts to authorities or emergency contacts in high-threat situations.

This project envisions empowering women to make informed decisions before traveling, providing an extra layer of safety during commutes, and ensuring that in moments when manual communication is impossible, the system can act as a silent guardian. By merging Ml, data analytics, and geolocation services, the Ml-Powered Women Safety & Emergency System represents a step towards building a safer environment for women and vulnerable individuals worldwide.

Introduction:-

Safety is a fundamental human right, yet women around the world continue to face significant threats to their security. From street harassment to violent crimes, incidents occur across urban and rural areas, often leaving victims without timely help or effective preventive measures. While mobile safety apps exist, most of them are reactive — they function only after a distress signal is manually triggered. In high-stress situations, such manual triggers may be impractical or impossible.

The need for a predictive and proactive safety system is greater than ever. Modern technologies such as Artificial Intelligence, Machine Learning, Natural Language Processing (NLP), and Geospatial Data Analysis offer the capability to predict risks, identify patterns, and act automatically in threatening circumstances. Leveraging these tools, safety systems can move beyond basic panic buttons to intelligent, context-aware guardians.

The Ml-Powered Women Safety & Emergency System combines multiple advanced modules into a single, user-friendly platform. Its Predictive Location Risk Analysis gives instant insights into the safety level of a location based on crime history, time of day, and police station proximity. The Crime Analysis by Age Group enables families to make informed decisions for the safety of children, teenagers, and elderly women. Through Real-Time GPS Tracking and Safe Route Suggestion, the system proactively guides users away from potential danger zones, functioning as a security-focused navigation assistant.

A standout feature, Intelligent Voice-Based Threat Detection, continuously monitors ambient audio to identify distress even when the victim cannot speak openly or use their phone. The integration of speech emotion recognition, gender detection, and keyword monitoring, combined with the user's current location data, makes it possible to automatically detect threats and send alerts.

By integrating predictive analytics with real-time monitoring and automatic emergency response, this system Mlms to bridge the gap between prevention and reaction, ensuring that women have a reliable, intelligent ally in safeguarding their journeys and daily activities.

Problem Statement:-

Women’s safety remains a global challenge despite the proliferation of smartphones, GPS technology, and social media connectivity. The existing safety applications primarily rely on manual intervention - pressing a panic button, sending an SOS message, or calling an emergency contact. In real-world scenarios, especially during sudden or high-intensity threats, victims may not have the time, physical ability, or presence of mind to manually activate these features. This delay in alerting can lead to loss of critical response time, potentially worsening the outcome.

Moreover, most safety tools do not provide preventive insights. A user stepping into an unfamiliar neighborhood has no reliable way to assess how risky that area is before arriving. Without predictive risk assessment, individuals often unknowingly enter high-crime areas. Similarly, there is minimal awareness of crime trends based on demographics, such as the vulnerability of certain age groups in particular regions, which could help families plan safer travel for minors and elderly women.

Another major gap is the lack of real-time adaptive navigation. Standard mapping applications may suggest the fastest route without considering safety factors such as crime hotspots, poor lighting, or lack of surveillance. There is a need for a navigation system that dynamically adjusts routes based on both safety and convenience.

Finally, voice-based passive threat detection is almost non-existent in public safety applications. A system that can interpret distress from background audio,detecting fear, anger, or keywords associated with danger - could potentially save lives, especially when the victim cannot manually interact with their phone. The absence of such an autonomous safety measure represents a critical vulnerability in existing solutions.

Proposed Solution:-

The Ml-Powered Women Safety & Emergency System is designed as a comprehensive safety platform that combines predictive analytics, real-time monitoring, intelligent voice analysis, and automated emergency response to provide proactive and reactive protection for women and vulnerable individuals.

1. Predictive Location Risk Analysis

- Function: Instantly evaluates the risk level of any entered location.
- Method: Uses historical crime datasets, proximity to police stations, and time-based crime trends to assess danger levels.
- Output: Displays a visual risk indicator — Low, Moderate, or High — to guide decision-making.
- Impact: Helps users avoid high-risk areas before traveling.

2. Crime Analysis by Age Group

- Function: Breaks down location-specific crime trends by age category (minors, young women, elderly).
- Method: Applies demographic-based data filtering and pattern recognition.
- Output: Shows statistical likelihood of crimes against specific age groups in an area.
- Impact: Empowers parents and guardians to choose safer environments for vulnerable family members.

3. Real-Time GPS Tracking + Safe Route Suggestion

- Function: Tracks user location and dynamically suggests safer routes.
- Method: Integrates live GPS data with mapped crime zones to reroute users away from danger.
- Output: A map interface highlighting safe paths in real time.
- Impact: Reduces exposure to high-risk areas during commutes or rideshares.

4. Intelligent Voice-Based Threat Detection

- Function: Detects distress through passive audio monitoring every 20 seconds.
- Modules:
  - Voice Gender Analysis – Identifies if the speaker is female.
  - Speech Emotion Recognition – Detects emotions like anger, fear, or distress.
  - Speech-to-Text Conversion – Converts conversations into text for keyword scanning.
  - Speaker Diarization – Determines the number of people and overlapping speech.
  - Threat Detection – Cross-checks keywords and emotions with high-crime location data to confirm potential danger.
  - Location-Based Alerting – Sends emergency alerts if distress is confirmed in a crime-prone area.
- Impact: Provides a hands-free, discreet emergency trigger when the victim cannot act manually.

Overall System Workflow:-

1. User enters or is detected in a location → System evaluates safety score.
2. User starts a journey → GPS tracking activates and safe routes are calculated.
3. Ambient audio monitoring → Voice and emotion analysis runs silently.
4. Potential threat detected → Immediate alert sent to authorities and emergency contacts with location details.

The proposed solution is not just a reactionary panic app but a multi-faceted Ml safety assistant that anticipates danger, continuously monitors for threats, and responds instantly — effectively bridging the gap between prevention and emergency action.

Methodology :-

The methodology for developing the Ml-Powered Women Safety & Emergency System follows a modular and data-driven approach, combining predictive analytics, real-time tracking, and Ml-based audio analysis. The implementation steps are:

1. Data Collection and Preprocessing

- Historical Crime Data – Collected from public safety databases, government crime statistics portals, and open data APIs.
- Location Data – Includes GPS coordinates, distances from police stations, and mapping of high-crime zones.
- Audio Datasets – Includes gender-labeled speech samples, emotion-labeled voice datasets, and distress keyword lists.
- Data Cleaning & Normalization – Ensures data accuracy, removes duplicates, fills missing values, and standardizes formats for analysis.

2. Predictive Location Risk Analysis

- Uses Machine Learning models such as Random Forest or Gradient Boosting to classify locations into Low, Moderate, High risk categories.
- Factors include:
  - Crime frequency in the area
  - Type of crimes reported
  - Time of day and seasonal variations
  - Distance from the nearest police station

3. Crime Analysis by Age Group

- Uses demographic filtering on crime datasets to detect age-specific crime trends.
- Outputs a visual breakdown of crime distribution by minors, young women, and elderly women for a given location.

4. Real-Time GPS Tracking + Safe Route Suggestion

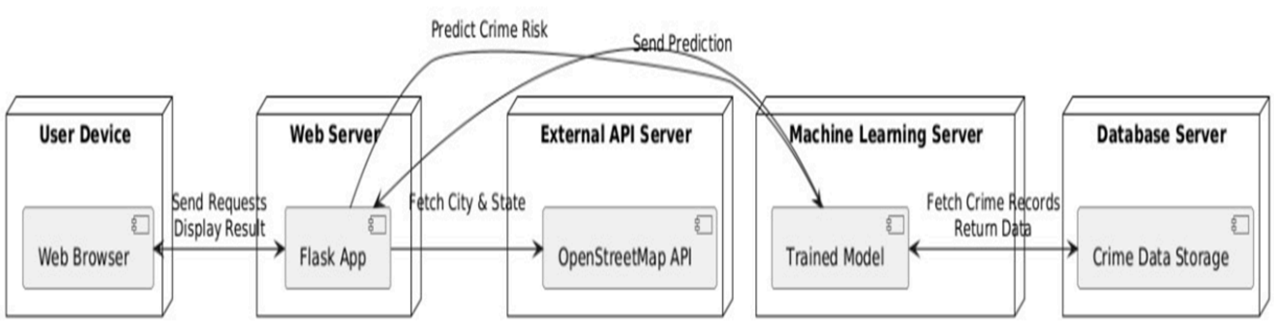
- GPS Module tracks user's live location.
- Routing Algorithm integrates with mapping APIs (Google Maps API / OpenStreetMap) and overlays crime heatmaps.
- Suggests safer alternate routes dynamically during travel.

5. Intelligent Voice-Based Threat Detection

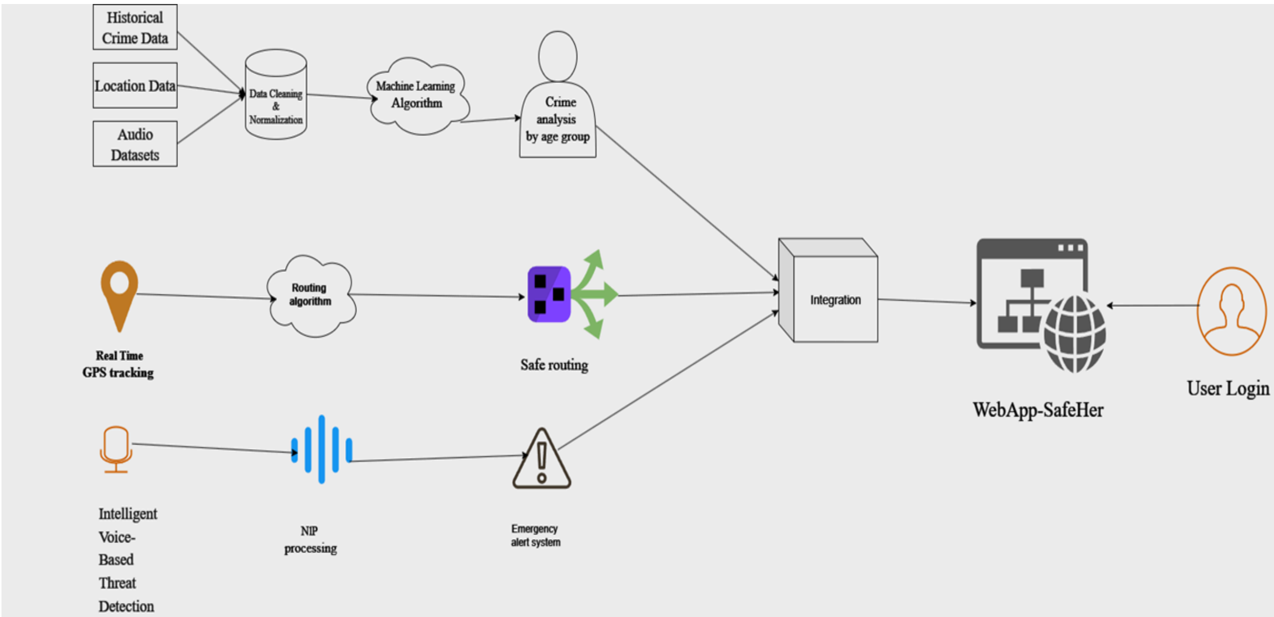
- Audio Processing: Captures ambient sound every 20 seconds.
- Gender Detection: Classifies voice as male/female using pre-trained deep learning models.
- Emotion Recognition: Identifies stress or distress through spectral and prosodic features.
- Speech-to-Text: Converts voice to text for keyword scanning.
- Threat Detection Algorithm: Correlates emotional tone, gender, and high-crime location to detect danger.
- Automatic Alerts: Sends emergency alerts to authorities and pre-defined contacts with live location.

6. Integration & Testing

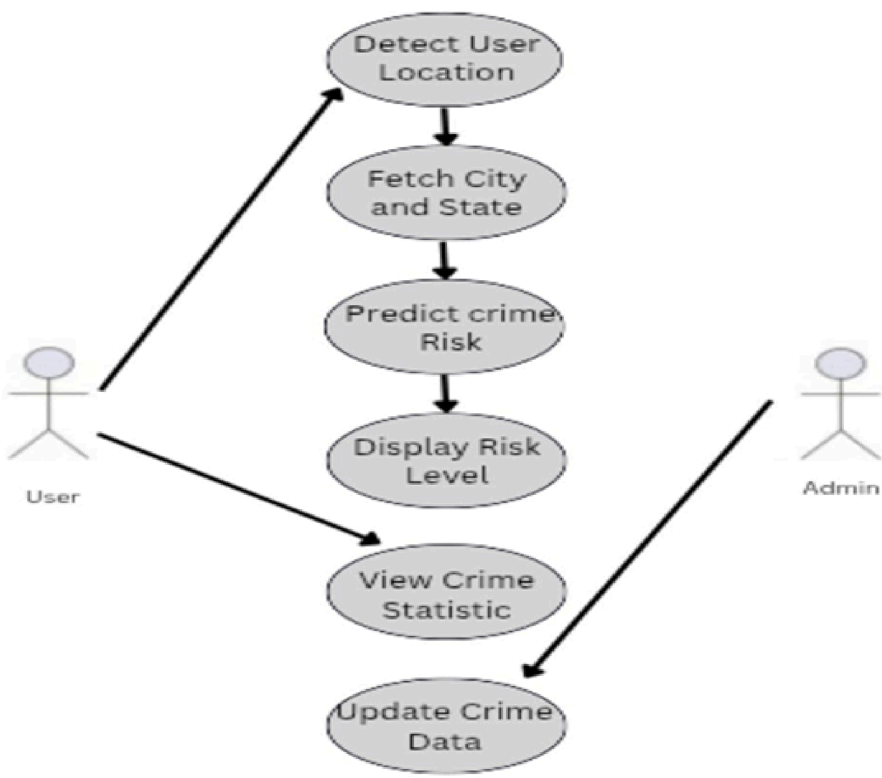
- Integrates all modules into a mobile or web application.
- Conducts unit testing for individual features and system testing for combined performance.



System Architecture:-



Use Case Diagram:



-

Hardware Requirements:-

Laptop/PC

Software Requirements:-

Category	**Technology / Tool**
Programming Language	Python (ML models, backend), React.Js (frontend)
Web Framework	Node.js,Express.js
Database	NoSQl
APIs	Google Maps API, Fast API
Machine Learning and Natural Language Processing	Pandas, Numpy, scikit-learn, TensorFlow, PyTorch
Audio Processing	Librosa, SpeechRecognition, PyDub
Version Control	Git, GitHub
Data Visualization	Matplotlib, Seaborn, PowerBI

Proposed Evaluation Measures :

To ensure the effectiveness, accuracy, and reliability of the ML-Powered Women Safety & Emergency System, the following evaluation measures will be applied:

1. Risk Analysis Accuracy – Evaluate predictive safety scores (Low/Moderate/High) using Precision, Recall, and F1-score against historical crime data.
2. Age-Based Crime Insights – Validate demographic crime patterns against official records for accuracy.
3. Safe Route Effectiveness – Measure how often suggested routes avoid crime zones without major travel delays.
4. Threat Detection Performance – Assess gender, emotion, and keyword detection accuracy along with false alarm rates.
5. Response Time – Track average time from threat detection to emergency alert dispatch (target < 5 seconds).
6. User Feedback – Gather usability ratings and trust levels from beta testers via surveys.

These measures will ensure the system is tested both technically and practically, covering accuracy, speed, reliability, and user satisfaction. A balanced evaluation will help identify strengths, address weaknesses, and guide improvements for real-world deployment.

Conclusion :

The ML-Powered Women Safety & Emergency System offers a proactive and intelligent approach to personal safety by combining predictive analytics, real-time GPS tracking, safe route suggestions, and voice-based threat detection into a unified platform. Unlike traditional safety applications that depend on manual intervention, this system anticipates potential threats, monitors surroundings continuously, and responds autonomously in critical situations. By leveraging machine learning, geospatial analysis, and advanced audio processing, it enables location-based risk assessment, age-specific crime insights, and hands-free emergency alerts, significantly reducing response time and enhancing user security. With further refinement, broader data integration, and collaboration with law enforcement, this solution can serve as a scalable, reliable safety companion for women and vulnerable individuals worldwide.

References:

1.Choudhary, A., Jindal, A., & Singh, P. (2021). Machine Learning Techniques for Crime Prediction and Classification: A Comparative Study. International Journal of Computer Applications.

2.Kumar, S., & Rani, M. (2022). A Review on Women Safety Apps and Crime Mapping Tools. Journal of Emerging Technologies and Innovative Research (JETIR).

3.Patel, R., & Shah, H. (2021). Crime Analysis using Machine Learning Algorithms. International Research Journal of Engineering and Technology (IRJET).

4.Sharma, V., & Jain, K. (2022). Smart Women Safety System Using GPS and Android App. International Journal of Scientific Research in Computer Science and Engineering.

5.Prakash, M., & Anjali, M. (2023). Crime Rate Analysis in India Using Machine Learning. Journal of Advanced Research in Dynamical and Control Systems.